

6. Serpenguzel, A., Arnold, S. & Griffl, G. Excitation of resonances of microspheres on an optical fiber. *Opt. Lett.* **20**, 654–656 (1995).
7. Chang, R. K. & Campillo, A. J. (eds) *Optical Processes in Microcavities* (World Scientific, Singapore, 1996).
8. Treussart, F. *et al.* Evidence for intrinsic Kerr bistability of high-Q microsphere resonators in superfluid helium. *Eur. Phys. J. D* **1**, 235–238 (1998).
9. Spillane, S. M., Kippenberg, T. J. & Vahala, K. J. Ultralow-threshold Raman laser using a spherical dielectric microcavity. *Nature* **415**, 621–623 (2002).
10. Campillo, A. J., Eversole, J. D. & Lin, H.-B. Cavity quantum electrodynamic enhancement of stimulated emission in microdroplets. *Phys. Rev. Lett.* **67**, 437–440 (1991).
11. Braginsky, V. B., Gorodetsky, M. L. & Ilchenko, V. S. Quality-factor and nonlinear properties of optical whispering-gallery modes. *Phys. Lett. A* **137**, 393–397 (1989).
12. Collot, L., Lefevre-Seguin, V., Brune, M., Raimond, J. M. & Haroche, S. Very high-Q whispering-gallery mode resonances observed on fused-silica microspheres. *Europhys. Lett.* **23**, 327–334 (1993).
13. Vernooy, D. W., Ilchenko, V. S., Mabuchi, H., Steed, E. W. & Kimble, H. J. High-Q measurements of fused-silica microspheres in the near infrared. *Opt. Lett.* **23**, 247–249 (1998).
14. Himeno, A., Kato, K. & Miya, T. Silica-based planar lightwave circuits. *IEEE J. Sel. Top. Quant.* **4**, 913–924 (1998).
15. Delfino, M. & Reifsteck, T. A. Laser activated flow of phosphosilicate glass in integrated circuit devices. *Elect. Device Lett.* **3**, 116–118 (1982).
16. McLachlan, A. D. & Meyer, F. P. Temperature dependence of the extinction coefficient of fused silica for CO₂ laser wavelengths. *Appl. Opt.* **26**, 1728–1731 (1987).
17. Sheik-bahae, M. & Kwok, H. S. Controlled CO₂ laser melting of silicon. *J. Appl. Phys.* **63**, 518–524 (1988).
18. Knight, J. C., Cheung, G., Jacques, F. & Birks, T. A. Phase-matched excitation of whispering-gallery-mode resonances by a fiber taper. *Opt. Lett.* **22**, 1129–1131 (1997).
19. Cai, M., Painter, O. & Vahala, K. J. Observation of critical coupling in a fiber taper to a silica-microsphere whispering-gallery mode system. *Phys. Rev. Lett.* **85**, 74–77 (2000).
20. Michler, P. *et al.* Laser emission from quantum dots in microdisk structures. *Appl. Phys. Lett.* **77**, 184–186 (2001).
21. Gayral, B. *et al.* High-Q wet-etched GaAs microdisks containing InAs quantum boxes. *Appl. Phys. Lett.* **75**, 1908–1910 (1999).
22. Kippenberg, T. J., Spillane, S. M. & Vahala, K. J. Modal coupling in traveling-wave resonators. *Opt. Lett.* **27**, 1669–1671 (2002).
23. Hansel, W., Hommelhoff, P., Hansch, T. W. & Reichel, J. Bose–Einstein condensation on a microelectronic chip. *Nature* **413**, 498–501 (2001).

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Logarithmic rate dependence of force networks in sheared granular materials

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Many models of slow, dense granular flows^{1–5} assume that the internal stresses are independent of the shearing rate. In contrast, logarithmic rate dependence is found in solid-on-solid friction^{6–8}, geological settings^{9–11} and elsewhere^{12–15}. Here we investigate the rate dependence of stress in a slowly sheared two-dimensional system of photoelastic disks, in which we are able to determine forces on the granular scale. We find that the mean (time-averaged) stress displays a logarithmic dependence on the shear rate for plastic (irreversible) deformations. However, there is no perceivable dependence on the driving rate for elastic (reversible) deformations, such as those that occur under moderate repetitive compression. Increasing the shearing rate leads to an increase in the strength of the force network and stress fluctuations. Qualitatively, this behaviour resembles the changes associated with an increase in density. Increases in the shearing rate also lead to qualitative changes in the distributions of stress build-up and relaxation events. If shearing is suddenly stopped, stress

relaxations occur with a logarithmic functional form over long timescales. This slow collective relaxation of the stress network provides a mechanism for rate-dependent strengthening.

Slow granular flows are typically described in the context of Mohr–Coulomb friction models² that resemble those used for describing friction between two solid bodies¹⁶. In the well-known solid friction scheme^{17,18}, an object on a frictional surface will resist a force and remain at rest, provided that the magnitude of the tangential force is less than the product of a static friction coefficient and the normal force. This picture was translated into the granular context (for dense granular systems characterized by networks of force chains¹⁹) by Coulomb²⁰ and more recent authors^{1,2}. In this context the normal and tangential forces are replaced by corresponding normal and shear stresses, and the surface of interaction is replaced by a plane within the material. For large enough tangential force relative to the normal force sliding friction (failure in a granular material) occurs. In these pictures, sliding friction (deformation following failure) is independent of the speed of sliding (the shear rate).

In reality, experiments in diverse contexts have shown that solid friction exhibits a logarithmic dependence on rate and that static frictional contacts strengthen logarithmically with age. These experiments span a vast range of lengths, and include studies at the atomic¹⁵, laboratory^{7,8,12–14} and geological scale^{9–11}. Recent experiments⁶ using granular materials sliding against the interior wall of a piston showed a clear rate dependence that was associated with ageing effects of individual solid-friction contacts and with the force network. Experiments^{7,8} in which a solid rough surface was pushed across a granular bed showed a slow strengthening (or ageing) with time of the (quasi-static) force, and creep before slip events. Other experiments¹¹ have found strengthening due to ageing for granular samples under large compressions. In the present experiments, which consider the grains and the force network, we show that there is a logarithmic rate-dependence in slowly sheared granular materials that is associated with irreversible rearrangements of the force network within the material itself. This is manifested as a strengthening with rate.

The experiments described here were carried out with a two-dimensional realization of a granular system. The particles were made of a photoelastic material, and were either relatively thin disks or flat particles with a pentagonal cross-section. By using a photoelastic material, we could determine forces at the grain scale. More detailed descriptions of the experimental apparatus and methods²¹

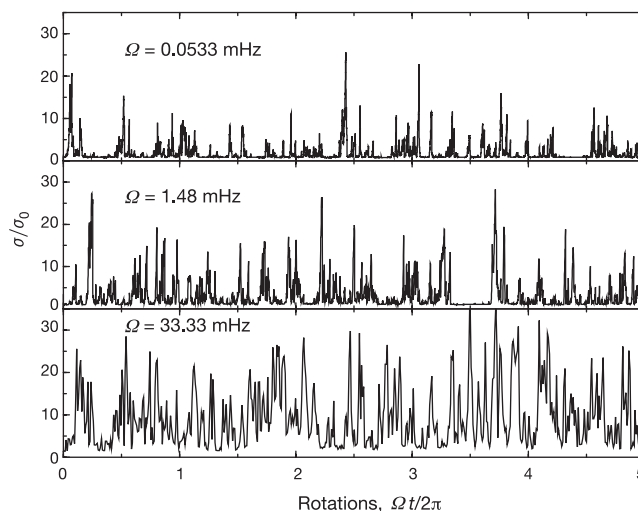


Figure 1 Time series for stress (normalized by $\sigma_0 \approx 4.11 \text{ N m}^{-1}$) for a range of shearing rates Ω that span the experimental range. These data are for pentagonal particles, but data for disks are qualitatively similar. Shear wheel radius 13.8 cm, $\gamma = \gamma_c + 0.0152$.

have been given elsewhere for related experiments^{19,22,23}, and here we provide only essential information for two qualitatively different experiments. In the first of these two experiments, the photoelastic particles were sheared in an annular geometry (Fig. 2 inset) that was bounded on the inside by a rough wheel, the source of shearing, and on the outside by an equally rough static ring. The particles sat on a horizontal, flat powder-lubricated Perspex sheet, and were contained from above by a similar sheet. This assembly was placed in a polariscope consisting of a pair of right and left circular polarizers, a light source and a video camera. Photoelastic images (effective shutter speed 10^{-2} s) were obtained at rates up to 15 Hz, captured by a frame-grabber, and processed in real time to determine the forces on the particles within the field of view of the camera (~ 200 particles). We computed the force on each particle in each frame and summed over all particles in a frame. This integrated force was then stored for each image to produce a time series over very long sampling times, between 2^{14} – 2^{17} points depending on the rate of shearing, Ω , and for 10 to approximately 200 rotations of the shearing wheel, depending on Ω . To a reasonable approximation ($\sim 10\%$), this quantity is proportional to the pressure, and for simplicity, we will refer to it as stress.

Important control parameters are Ω ($=2\pi/T$, where T is the period for one rotation of the shearing wheel) and the density of the system, which we give in terms of the area fraction, γ , occupied by the particles. As shown previously^{19,22}, there is a critical value when $\gamma = \gamma_c$, below which the force network vanishes, and we typically reference γ to this value. (γ_c depends on properties such as the particle shape and possibly the geometry of the container.) It is also useful to give the rotation rate, $f = 1/T = \Omega/2\pi$. In these experiments they were $0.03 \text{ MHz} \leq f \leq 60 \text{ MHz}$.

In Fig. 1 we show several examples of stress time series for values of Ω ranging over about three orders of magnitude. The three images of this figure give an indication of the effect of the shearing rate; the time series for the slowest rate is clearly of lower amplitude and more intermittent than that for the highest rate.

The rate dependence is seen clearly in the mean (time-averaged)

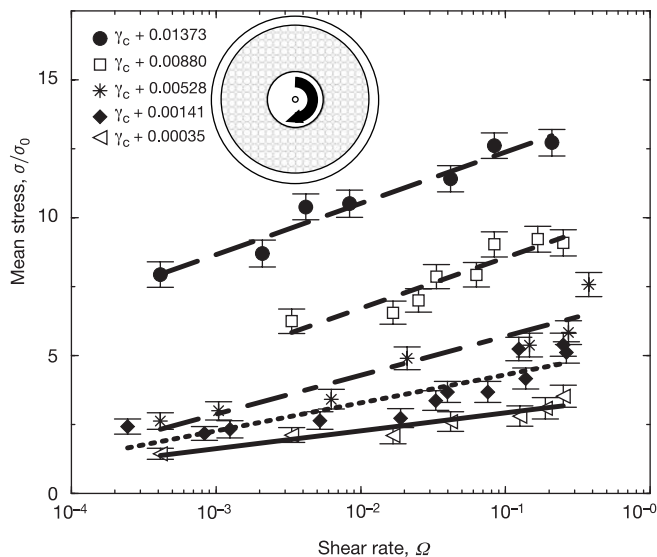


Figure 2 Mean stress versus shearing rate, Ω , for a range of densities for pentagonal particles. Note that these densities are relatively close to the critical value (γ_c) at which the force network vanishes. The linear-logarithmic scales emphasize the fact that the data are consistent with a logarithmic variation of the stress with Ω . Here, $\sigma_0 \approx 4.11 \text{ N m}^{-1}$. Inset, sketch of Couette shear apparatus. The particles used in the experiment had typical linear dimensions between 0.7 cm and 0.9 cm, a Young's modulus of 4 MPa and a density 1.15 g cm^{-3} . For either shear or compression experiments, the humidity varied by no more than $\pm 5\%$.

stress versus Ω in Fig. 2. These data are consistent with a logarithmic rate dependence for all values of γ . Similar effects have been seen in previous experiments^{6,8,11,12} but with a significant difference: in the present experiments, there is a logarithmic growth of the mean stress (that is, the force network) with rate, while other experiments show a logarithmic strengthening of the friction coefficient with age and/or humidity. If we fit these data to the form $\bar{\sigma} = A \log[\Omega/\Omega_0]$, then the amplitude, A , is sensitive to the density, but the characteristic frequency, Ω_0 , depends only weakly on γ , and corresponds to a typical time of $2\pi/\Omega_0 \approx 30 \text{ d}$. However, there is no compelling reason to believe that this relation holds in the limit $\Omega \rightarrow 0$. A more realistic description may be $\bar{\sigma} = A \log[(\Omega + \Omega_1)/\Omega_0]$, where $\Omega_1 > \Omega_0$.

If a sample is sheared under steady-state conditions and the shearing wheel is abruptly stopped, the force network relaxes over very long timescales. In Fig. 3a, we show the changes that have occurred in a particular case after 10 h. In Fig. 3b we show quantitative data for the relaxation of stress versus time on a semi-log plot. Note that there are two timescales: a quicker initial decay lasting about 20–300 s and a much longer process such that the stress network may still be relaxing after 20 h. This behaviour may provide insight into the creep events observed⁷ in granular friction experiments. Figure 3b shows fits to the stress decrease at long times of the form $\sigma(t)/\sigma(t=0) = B \log(t/t_0)$. Although the relaxation is not completely uniform in time, a logarithmic functional form is not unreasonable for many cases (after an initial rapid relaxation). Note an example of a major relaxation event in one of these runs about 350 s after the cessation of shearing: thereafter, continued slow relaxation occurred. In Fig. 3c we contrast relaxation stress data for samples prepared under uniform compression to the relaxation of initially sheared samples in Fig. 3b. In the compression experiments no relaxation is seen within experimental resolution.

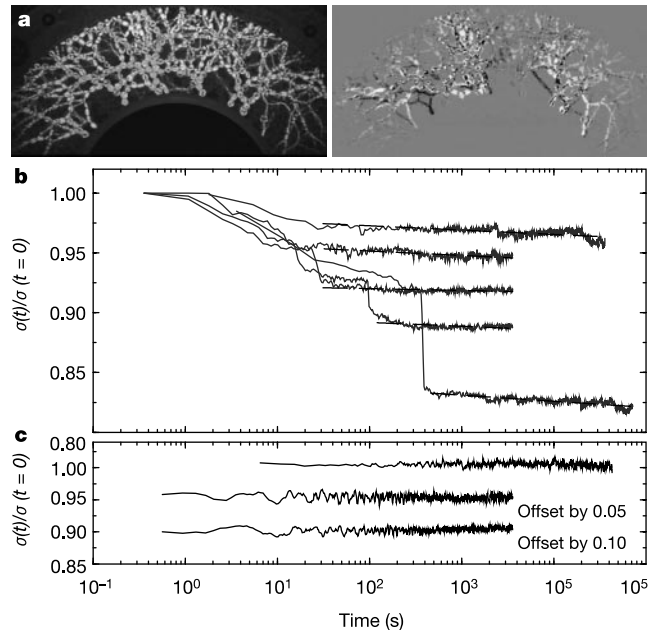


Figure 3 Stress relaxation in sheared and compressed systems. **a**, For the shear experiment, we show the initial force network (left) and the change in force network over a 10-h period (right). In the latter image, decreases and increases in the photoelastic image intensity are shown in white and black, respectively. **b**, Relaxation for particles that were undergoing plastic shear deformation due to shear before $t = 0$. At $t = 0$, the shearing wheel was stopped and maintained in a fixed position. The dashed lines are a fit to $\sigma(t)/\sigma(t=0) = B \log(t/t_0)$. **c**, Similar stress versus time data for a collection of particles that are maintained under uniform compression.

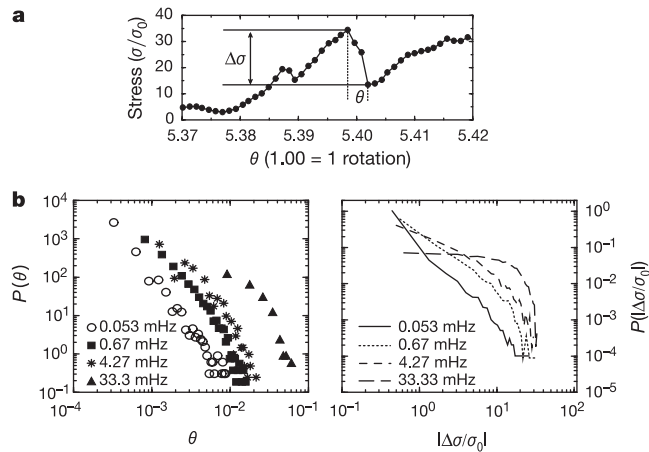


Figure 4 Stress “avalanches” in sheared granular materials. Here, $\sigma_0 \approx 4.11 \text{ N m}^{-1}$; particles were pentagons with $\gamma = \gamma_c + 0.022$. **a**, A short segment of a time series that identifies a single avalanche, $\Delta\sigma$ and θ . **b**, Distributions $P(\theta)$ and $P(|\Delta\sigma/\sigma_0|)$ of stress avalanches for a dense system of pentagons for several shear rates.

We gain additional insight into the effect of rate changes by determining the distribution of build-up events (monotonic increases in stress) and release events (‘avalanches’—monotonic decreases in stress). Here, there are two aspects that are of interest: the size of the build-ups and avalanches, $\Delta\sigma$, and the angle, θ , through which the shearing wheel turns during the course of a build-up or an avalanche. In Fig. 4a, we show an example of an avalanche event, and in Fig. 4b, we show distributions of $\Delta\sigma$ and θ for avalanches. For near-critical γ and for the slowest Ω the avalanche distributions are consistent with the power laws. At higher γ and Ω the distributions for both build-up (not shown) and avalanche events are roughly exponential. We note that increases in both Ω and in γ tend to have the same qualitative effect on the statistics of avalanches and build-ups.

To further explore the origin of this rate dependence, we have carried out a second set of experiments, using the same photoelastic techniques, in which the particles were confined to a box with three fixed sides. The sample was then compressed and released repeatedly on the fourth side by means of an oscillating piston (Fig. 5 inset). The driving amplitude of the piston ($a = 0.44 \text{ cm}$) was chosen so that the typical piston velocity matched the corresponding velocities of the shear wheel in the first experiment. In the first set of experiments there was continuous plastic shear deformation, whereas there was none in the second set—the particles maintained their relative position. Within experimental error, there was no rate dependence of this process (Fig. 5 main figure).

We have demonstrated that sheared granular materials exhibit a logarithmic rate dependence that is tied in an essential way to force network/particle rearrangements. The nature of this rate dependence is further illuminated by the slow relaxation that occurs when shearing is halted. These experiments in particular show that there are slow collective rearrangements of the force network that can occur over large timescales. Although the frictional micro-contacts between particles may be strengthening in time^{8,12,16}, the force network as a whole is relaxing over very long timescales. This slow relaxation of forces is consistent with increases in stress with rate. Force chains generated by shearing cannot completely relax on the timescales over which new chains are formed, an effect that becomes stronger with increasing Ω . It seems likely that this effect is related to the long timescales associated with collective rearrangements in other settings, such as compaction in granular materials²⁴. For compaction, rearrangements become progressively more difficult over time, because each new rearrangement requires the involvement of larger collections of particles. Similarly, in the

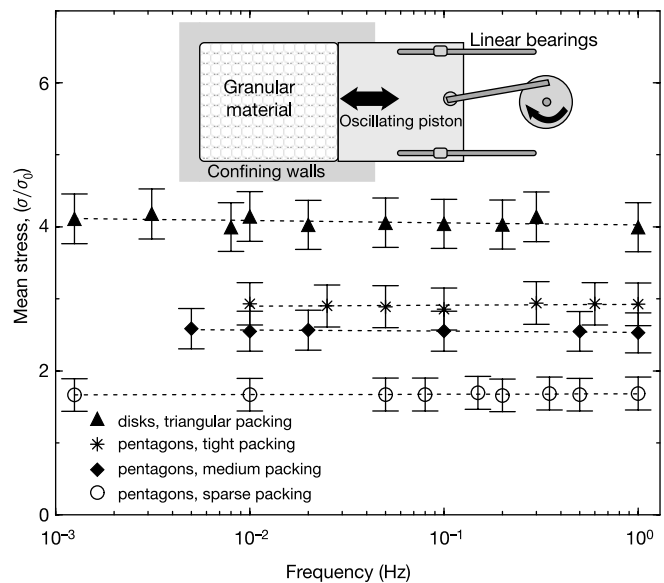


Figure 5 Mean stress versus oscillation rate, ω , for several densities for oscillatory compression. Data for both disks and pentagons are shown. The linear-logarithmic scales emphasize the fact that there is no variation of the stress with ω within experimental uncertainties. $\sigma_0 \approx 3.83 \text{ N m}^{-1}$. Inset, sketch of piston apparatus.

present experiments, relaxation events involving collections of particles become less probable over time owing to geometric constraints on the particles, to the fact that those contacts that are near failure will have slipped, and to the fact that the available elastic energy is gradually reduced with each rearrangement. An interesting question is whether similar properties could be found in other jammed systems²⁵ such as colloids or foams. □

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- Wood, D. M. *Soil Behaviour and Critical State Soil Mechanics* (Cambridge Univ. Press, Cambridge, UK, 1990).
- Nedderman, R. M. *Statics and Kinematics of Granular Materials* (Cambridge Univ. Press, Cambridge, UK, 1992).
- Jaeger, H. M., Nagel, S. R. & Behringer, R. P. Granular solids, liquids, and gases. *Rev. Mod. Phys.* **68**, 1259–1273 (1996).
- Herrmann, H. J., Hovi, J. P. & Luding, S. (eds) *Physics of Dry Granular Media* (NATO ASI Series, Kluwer, Dordrecht, 1997).
- Behringer, R. P. & Jenkins, J. T. (eds) *Powders and Grains 97* (Balkema, Rotterdam, 1997).
- Overez, G., Kolb, E. & Clément, E. Rheology of a confined granular material. *Phys. Rev. E* **64**, 060302–060305(R) (2001).
- Nasuno, S., Kudrolli, A., Bak, A. & Gollub, J. P. Time-resolved studies of stick-slip friction in sheared granular layers. *Phys. Rev. E* **58**, 2161–2171 (1998).
- Losert, W., Gémard, J.-C., Nasuno, S. & Gollub, J. P. Mechanisms for slow strengthening in granular materials. *Phys. Rev. E* **61**, 4060–4068 (2000).
- Dieterich, J. H. Modeling of rock friction 1: Experimental results and constitutive equations. *J. Geophys. Res.* **84**, 2161–2168 (1979).
- Ruina, A. L. Slip instability and state variable friction laws. *J. Geophys. Res.* **88**, 10359–10370 (1983).
- Karner, S. L. & Marone, C. The effect of shear load on frictional healing in simulated fault gouge. *Geophys. Res. Lett.* **25**, 4561–4564 (1998).
- Berthoud, P., Baumberger, T., G'Sell, C. & Hiver, J. M. Physical analysis of the state- and rate-dependent friction law: Static friction. *Phys. Rev. B* **59**, 14313–14327 (1999).
- Baumberger, T., Berthoud, P. & Caroli, C. Physical analysis of the state- and rate-dependent friction law. II. Dynamic friction. *Phys. Rev. B* **60**, 3928–3939 (1999).
- Carlson, J. M. & Batista, A. A. Constitutive relation for the friction between lubricated surfaces. *Phys. Rev. E* **53**, 4153–4165 (1996).
- Gnecco, E. et al. Velocity dependence of atomic friction. *Phys. Rev. Lett.* **84**, 1172–1175 (2000).
- Heslot, F., Baumberger, T., Perrin, B., Caroli, B. & Caroli, C. Creep, stick-slip, and dry-friction dynamics: Experiments and a heuristic model. *Phys. Rev. E* **49**, 4973–4988 (1994).
- Persson, B. N. J. *Sliding Friction: Physical Principles and Applications* (Springer, Berlin, 1998).
- Meyer, E., Overney, R. M., Dransfeld, K. & Gyalog, T. *Nanoscience, Friction and Rheology on the Nanometer Scale* (World Scientific, Singapore, 1998).
- Howell, D., Veje, C. & Behringer, R. P. Stress fluctuations in a 2D granular Couette experiment: A continuous transition. *Phys. Rev. Lett.* **82**, 5241–5244 (1999).
- Coulomb, A. *Mémoires de Mathématiques et de Physique* (L'Imprimerie Royale, Paris, 1776).
- Heywood, R. B. *Designing by Photoelasticity* (Chapman and Hall, London, 1952).
- Howell, D. W., Behringer, R. P. & Veje, C. T. Fluctuations in granular media. *Chaos* **9**, 559–572 (1999).
- Geng, J. et al. Footprints in sand: The response of a granular material to local perturbations. *Phys. Rev. Lett.* **87**, 035506–035509 (2001).

24. Nowak, E. R., Knight, J. B., Ben-Naim, E., Jaeger, H. M. & Nagel, S. R. Density fluctuations in vibrated granular. *Phys. Rev. E* **57**, 1971–1982 (1998).
25. Liu, A. L. & Nagel, S. R. Nonlinear dynamics: Jamming is just not cool anymore. *Nature* **396**, 21–22 (1998).

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Early history of Earth's crust–mantle system inferred from hafnium isotopes in chondrites

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The ^{176}Lu to ^{176}Hf decay series has been widely used to understand the nature of Earth's early crust–mantle system^{1–6}. The interpretation, however, of Lu–Hf isotope data requires accurate knowledge of the radioactive decay constant of ^{176}Lu ($\lambda_{176\text{Lu}}$), as well as bulk-Earth reference parameters. A recent calibration of the $\lambda_{176\text{Lu}}$ value calls for the presence of highly unradiogenic hafnium in terrestrial zircons with ages greater than 3.9 Gyr, implying widespread continental crust extraction from an isotopically enriched mantle source more than 4.3 Gyr ago⁷, but does not provide evidence for a complementary depleted mantle reservoir. Here we report Lu–Hf isotope measurements of different Solar System objects including chondrites and basaltic eucrites. The chondrites define a Lu–Hf isochron with an initial $^{176}\text{Hf}/^{177}\text{Hf}$ ratio of 0.279628 ± 0.000047 , corresponding to $\lambda_{176\text{Lu}} = 1.983 \pm 0.033 \times 10^{-11} \text{ yr}^{-1}$ using an age of 4.56 Gyr for the chondrite-forming event. This $\lambda_{176\text{Lu}}$ value indicates that Earth's oldest minerals were derived from melts of a mantle source with a time-integrated history of depletion rather than enrichment⁷. The depletion event must have occurred no later than 320 Myr after planetary accretion, consistent with timing inferred from extinct radionuclides^{8–10}.

The main stage of Earth's accretion was completed approximately 30 Myr (refs 11, 12) after the condensation of the first solids from the early solar nebula $4,567.2 \pm 0.6$ Myr ago¹³. However, the exact timing of crust–mantle separation and onset of convection in the Earth's mantle remains poorly constrained, a critical consideration for the understanding of early planetary differentiation processes. Formation of either oceanic or continental crust results in chemical depletion, and therefore isotopic fractionation, of the residual mantle with time. Moreover, when crust is extracted, it records the isotopic composition of its source reservoir and thus provides information regarding the time-integrated history of either depletion or enrichment of its source. The Lu–Hf isotopic system, with a half-life of about 37 Gyr (ref. 7), is both a powerful chronometer and petrogenetic tool, allowing precise dating of geological materials as well as the identification of ancient Lu/Hf fractionation in planetary geochemical reservoirs.

Intrinsic to application of such isotopic tracers to the study of the early Earth is knowledge of chondritic (bulk Earth) reference values

for the Lu–Hf system (see Fig. 1) and the precise value of the decay constant for the parent isotope (^{176}Lu). A recent calibration⁷ of $\lambda_{176\text{Lu}}$ yielded a value of $1.865 \pm 0.015 \times 10^{-11} \text{ yr}^{-1}$, approximately 4% slower than previous estimates, but in agreement with two recent decay-counting experiments^{14,15}. This value has remained controversial, because the Hf isotopic composition of Earth's oldest minerals (≥ 4 -Gyr-old detrital zircons from the Jack Hills meta-sediments, Australia⁵), when back-calculated with this $\lambda_{176\text{Lu}}$ value, indicate derivation from isotopically enriched source reservoirs ($\varepsilon_{\text{Hf}} = -2$ to -6 between 3.97 and 4.14 Gyr ago; the ε_{Hf} value is the deviation of the $^{176}\text{Hf}/^{177}\text{Hf}$ ratio of a sample from the chondritic uniform reservoir—CHUR—in parts per 10^4). These unradiogenic Hf isotope signatures require that Earth had developed a first persistent crust of felsic nature by 4.3 Gyr ago, and questions the existence of a possible early mafic proto-crust¹⁶ because model ages for mafic protoliths are ~ 200 Myr older, that is, during the early stage of planetary accretion. In contrast, the evidence for a complementary reservoir of high $^{176}\text{Hf}/^{177}\text{Hf}$ values that would account for widespread pre-3.8-Gyr crustal production is lacking, because younger (~ 3.5 – 3.8 Gyr) juvenile lithologies from Greenland, South Africa and Australia seem to be derived from an unfractionated mantle source region (Fig. 2a), a feature that is not well reconciled with Nd isotope studies of early Archaean lithologies¹⁷.

Previous efforts to constrain the initial Hf isotope composition of the Solar System were either plagued by large analytical uncertainties^{18,19}, or based on the average present-day value for a number of carbonaceous and ordinary chondrites, which failed to define an isochron relationship²⁰. The latter approach is problematical, because it is highly dependent on precise knowledge of both the $^{176}\text{Lu}/^{177}\text{Hf}$ ratio for CHUR and the $\lambda_{176\text{Lu}}$ value. In order to accurately re-determine the initial $^{176}\text{Hf}/^{177}\text{Hf}$ ratio of the Solar

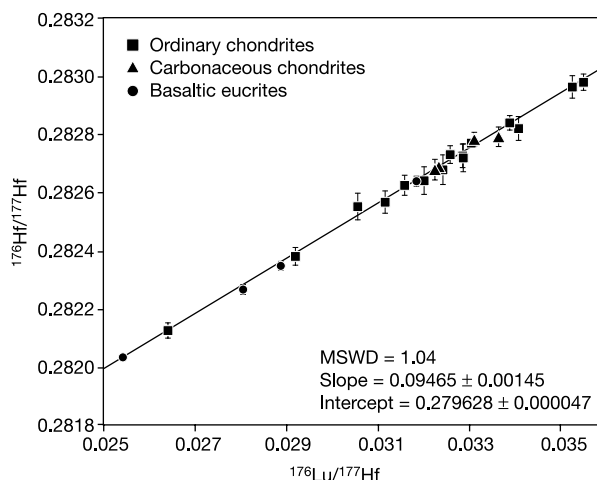


Figure 1 Lu–Hf isochron for chondrites analysed in this study. We interpret the intercept of the regression to represent the initial $^{176}\text{Hf}/^{177}\text{Hf}$ ratio of the Solar System. MSWD, mean square of weighted deviations. Errors on the $^{176}\text{Lu}/^{177}\text{Hf}$ ratios are 0.2% (2 s.d.) and are smaller than the plotting symbols. Error bars on the $^{176}\text{Hf}/^{177}\text{Hf}$ determinations represent 2 s.d. reproducibility, based on a relationship between in-run precision and external reproducibility of an in-house Hf standard. Excluding the carbonaceous chondrites yields a slope and intercept of 0.09465 ± 0.00147 and 0.279629 ± 0.000047 , respectively. Likewise, the L chondrites alone yield a slope and intercept of 0.09503 ± 0.00154 and 0.279619 ± 0.000049 . A recent comprehensive Lu–Hf study of basaltic eucrites yielded a slope and intercept of 0.0937 ± 0.0016 and 0.279638 ± 0.000046 , identical to the values determined here for chondrites (ref. 27; the eucrite Caldera, an impact melt, was excluded from the regression because Sm–Nd systematics indicate post-crystallization disturbance). All regressions were performed using isochron software available at ftp.ens-lyon.fr/pub/users/GEOL/albarede (written in Matlab, ref. 27). Data shown in this figure, together with details of the experimental methods used in their determination, are available as Supplementary Information.